

Answer **all** questions.

1. Cystic fibrosis (CF) is an inherited condition caused by a DNA mutation.

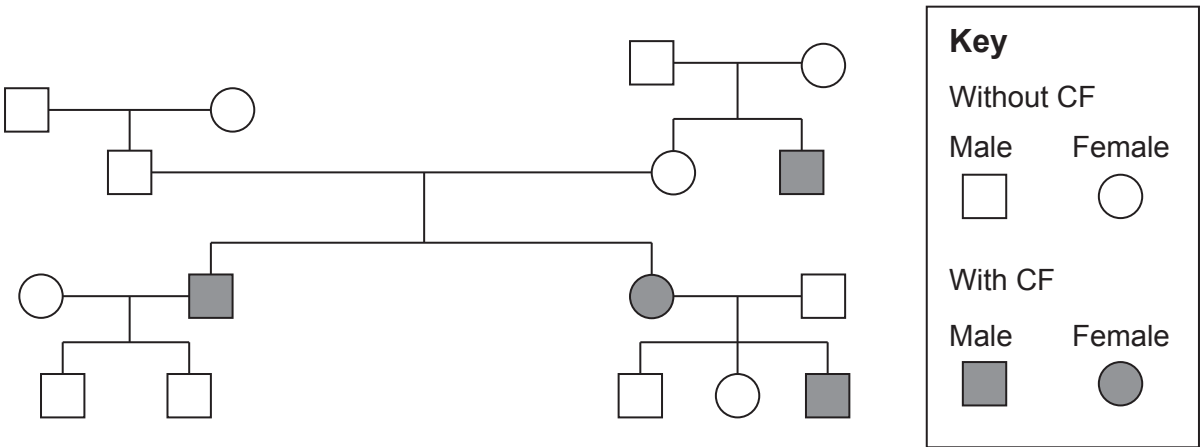
(a) **Complete the sentences** by selecting your answers from the words below. [3]

radiation	increase	random	prevent	regular
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A mutation is a change in DNA.

Ionising can the rate of mutations.

(b) In the family tree below, some people have CF.



(i) Calculate the **percentage of people** in this family tree who have CF. [2]

Percentage = %

(ii) State how many males and females in this family tree have CF. [1]

Males =

Females =



(iii) In the whole population of the UK,

- 0.01% of people have CF
- The ratio of males to females is 1 : 1

From your answers to (i) and (ii), give **two** ways in which this family is different from the whole population of the UK: [2]

I.

II.

(c) CF affects the lungs.

In a treatment, patients with CF are given DNA which does not have the CF mutation.

Complete the following sentences by underlining the correct word. [2]

(i) The treatment is called:

chemotherapy

gene therapy

physiotherapy

(ii) The DNA is given to the patient by:

injection

infusion

inhalation

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